



Overview



REMEDIATION REPORT
for
MEGAQUAGGA PUBLISHING



MQP

Version 1.0 2025-08-11

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Presented to **MEGAQUAGGA PUBLISHING**

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EXECUTIVE SUMMARY

This report outlines the remediation work completed for MegaQuagga Publishing to address security weaknesses identified in its wordpress environment. The primary issues involved exposure to remote file inclusion attacks and a lack of inspection on encrypted web traffic. To resolve these risks, ModSecurity was deployed and tuned as a web application firewall (WAF), and SSL offloading was configured on the apache reverse proxy. These changes ensure malicious requests are filtered while legitimate traffic continues to flow securely.



Actions Taken

ACTIONS TAKEN

This section outlines the actions taken on which assets to remediate which vulnerabilities.

Vulnerability	Remediation Solution
WordPress is vulnerable to web application attacks	Implemented mod security and WAF to block RFI
Unencrypted web traffic	Set up apache reverse proxy to handle SSL termination and enforce HTTP traffic

Remediation Solution #1: Used pfsense diagnostics to begin packet capture and observe initial traffic behavior.

Verified the original RFI exploit worked before remediation.

Installed and enabled mod security on the apache reverse proxy

Confirmed mod security blocked RFI attempts while allowing legitimate web traffic

Saved PCAP evidence showing malicious requests being successfully blocked.

Remediation Solution #2: Verified SSL certificates were available and trusted.

Created and enabled a virtual host configuration for HTTPS (port 443)

Restarted apache and confirmed the site was serving traffic securely over HTTPS

Captured a new PCAP showing encrypted traffic and inspected decrypted traffic at the proxy/WAF layer

Confirmed normal user traffic remained functional and malicious RFI attempts were still blocked.

☒ Evidence of Remediation

EVIDENCE OF REMEDIATION

This section documents evidence of the remediations being implemented. This evidence may take the form of details from packet captures (for example, alterations in traffic flow or packet headers), ModSecurity logs appearing in the Graylog SIEM, or output from command-line tools. For each remediation, we provide evidence of the original state as well as the current altered state.

Reverse proxy

These are alterations to MegaQuagga's environment associated with implementing the reverse proxy server responsible for implementing our two remediations (WAF & SSL offloading).

Item	Original Vulnerable State	Current Remediated State
SSL Offloading / HTTPS enforcement	Website traffic was being served over plain HTTP with no encryption.	The reverse proxy now terminates SSL and serves the site over HTTPS on port 443, ensuring traffic is encrypted.

Web Application Firewall (WAF)

These are alterations to MegaQuagga's environment associated with implementing the WAF remediation.

Item	Original Vulnerable State	Current Remediated State
RFI exploit	Remote File inclusion attempts were processed and succeeded, allowing unauthorized command execution.	Modsecurity is now active and configured to block RFI-style requests, returning a denial response when malicious patterns are detected.

SSL Offloading

These are alterations to MegaQuagga’s environment associated with implementing the WAF remediation.

Item	Original Vulnerable State	Current Remediated State
Encrypted Communication	Traffic between client and server traveled unencrypted, exposing data in transit,	SSL offloading ensures secure HTTPS communication, with certificates validated at the reverse proxy.

Recommendations

RECOMMENDATIONS

This section outlines recommendations for continued monitoring and next steps for future remediations to further strengthen MegaQuagga's security posture.

Recommendation #1: Log Review and monitoring

Continue reviewing WAF and reverse proxy logs on regular schedule to quickly detect unusual or suspicious web activity. This helps identify new attack attempts early and ensures the protections remain effective over time.

Recommendation #2: Maintain security configurations

Keep ModSecurity rules set, Wordpress plugins/themes, and SSL certificates up to date. Updating these components reduces exposure to newly discovered vulnerabilities and ensures encryption remains valid and trusted.

Recommendation #3: Administrative awareness

Provide brief training or guidance to site administrators on safe plugin usage, configuration practices, and recognizing indicators of malicious activity. Ensuring administrators know how misconfigurations happen helps prevent the same vulnerability from returning.



Conclusion

CONCLUSION

The vulnerabilities were resolved by enforcing HTTPS and deploying ModSecurity to block malicious requests. Packet captures confirmed the exploit no longer succeeds, and the environment is now more secure.

Tab 6

